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ORIGINAL ARTICLE

Various optimized machine learning techniques to predict agricultural commodity prices

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Abstract

Recent increases in global food demand have made this research and, therefore, the prediction of agricultural commodity prices, almost imperative. The aim of this paper is to build efficient artificial intelligence methods to effectively forecast commodity prices in light of these global events. Using three separate, well-structured models, the commodity prices of eleven major agricultural commodities that have recently caused crises around the world have been predicted. In achieving its objective, this paper proposes a novel forecasting model for agricultural commodity prices using the extreme learning machine technique optimized with the genetic algorithm. In predicting the eleven commodities, the proposed model, the extreme learning machine with the genetic algorithm, outperforms the model formed by the combination of long short-term memory with the genetic algorithm and the autoregressive integrated moving average model. Despite the fluctuations and changes in agricultural commodity prices in 2022, the extreme learning machine with the genetic algorithm model described in this study successfully predicts both qualitative and quantitative behavior in such a large number of commodities and over such a long period of time for the first time. It is expected that these predictions will provide benefits for the effective management, direction and, if necessary, restructuring of agricultural policies by providing food requirements that adapt to the dynamic structure of the countries.

Keywords Prediction · Commodity prices · Artificial neural network · Extreme Learning Machine · Long short-term memory · Genetic algorithm · Autoregressive integrated moving average

1 Introduction

Nowadays, the recovery process of countries continues after the economic shock caused by the COVID-19 epidemic. The prospect of economic normality and the decreasing impact of the pandemic globally have been fueling consumer demand, which has been driving price hikes. While price increases have caused inflation in almost every country today, the rise in food prices stands out as one of the primary causes of inflation. Furthermore, the world has currently been facing a food security crisis as well as supply chain tensions. Prices for agricultural commodities have already been significantly impacted by global climate change, and price swings will unavoidably be influenced by factors such localized floods and droughts [1]. On the other hand, this inflationary climate has been made worse by the huge rises in the price of electricity and fertilizer. The cost of farming rises, which raises the price of agricultural commodities. This is because agricultural machinery, vehicles, and buildings use more energy during the agricultural production process. In addition to upsetting food supply systems, COVID-19 is reducing food reserves and increasing food demand. As a result, the cost of food and agricultural commodities is increasing and reaching new highs. The Russia–Ukraine war additionally compound the threat to the global food supply chain by

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disrupting production, planting, harvesting, and shipping, in addition to climate disasters, pandemics, and rising production costs [2, 3]. This supply shock has an immediate effect on agricultural commodity prices, sending them to all-time highs. Food import and export restrictions also raise the overall price of food and its volatility [4]. As a result, between April 2020 and March 2022, food prices increased by 84 percent, which was the highest increase since 2008 [5]. Agricultural commodity price spikes that are excessively over forecasted levels are referred to as crises, and they lead to significant economic and societal issues. The strategies of political leaders may be in doubt if the price of a highly desired agricultural product increases dramatically, as happened when onion prices in India affected election outcomes [6] or when food prices in Algeria drove people onto the streets in 2011 [7]. When the price crises of various agricultural items, which first appeared in 1966, are analyzed, it is discovered that they have evolved in response to changing economic and political circumstances, such as the shift from agriculture to urban regions [8]. Furthermore, it has been determined that the incidence of localized crises has a limited worldwide influence. Production and trade disruptions caused worldwide agricultural commodity price crises in the 1970s [9]. Concerns about the volatility of agricultural commodity prices are now linked to the concept of food security. Factors such as the availability of agricultural products, economic and physical access to these products, and the consumption of agricultural products generate price changes and major issues in their supply [10]. With the recent addition of the Ukraine-Russia war, the difficulties experienced in physical access to many agricultural items as a result of global conflicts have had a substantial impact on agricultural commodity prices. Agricultural commodity prices are directly related to many factors such as weather conditions [11], climate changes [12], supply and demand factors [13] and international crises [14]. These factors play an important role in determining agricultural commodity prices. Many sectors such as investors, food companies and consumers wish to predict agricultural commodity prices. Therefore, it is believed that forecasting agricultural commodity prices will help those operating in this sector to make strategic planning and will play an important role in issues such as food security and economic stability. Forecasting is difficult due to the combination of many variables affecting agricultural commodity prices, many of which are based on uncontrollable natural factors. As a result, forecasting agricultural commodity prices emerges as a process characterized by inherent uncertainties, and achieving precision in forecasts proves to be a considerable challenge. Despite these challenges, applying accurate prediction models without considering the causal relationship between variables can help those operating in the sector make more informed decisions and be prepared for possible risks. As a result, it is a fact that forecasting the behavior and trends of agricultural commodity prices would enable the establishment of appropriate policies to address potential local or global problems that may arise. Many ways have been proposed in the literature to carry out this prediction [15, 16]. Many artificial intelligence models have lately been applied to anticipate the behavior and trends of agricultural commodity prices. With the application of these models, studies aimed at estimating agricultural commodity prices by considering variables such as product quality levels [17], previous prices [18], agricultural illnesses [19], seasonal factors [20] and climate change [21] come to the fore. Although many different models have yielded successful results, policymakers are concerned that the expected contributions may remain at relatively low levels due to the short-term nature of these predictions. In this study, three different models have been proposed with the help of artificial intelligence techniques to predict the price change behavior of eleven different agricultural items, including crops like wheat, corn, and soybeans, which have increased in price in recent years. Sugar, rice, oat, cotton, coffee, cocoa, soybean oil, and lumber came to the fore in this context in the grain corridor constructed in Turkey under the UN coordination amid the current global supply crisis. The first of the three models is the autoregressive integrated moving average (Auto-ARIMA) [22], which is a standard method for estimating agricultural commodity prices. One of the remaining two proposed models is the long short-term memory with genetic algorithm (GA-LSTM) model, which is created by merging genetic algorithm (GA) and long short-term memory (LSTM) [23]. In predicting long-term variations and trends in agricultural product commodity prices, the third proposed model, a hybrid of the genetic algorithm and the extreme learning machine (GA-ELM), is seen to outperform both the Auto-ARIMA and GA-LSTM models. Some important mathematical abbreviations used in this paper are listed in Table 1.

Several recent researches have focused on the construction and optimization of various machine learning models to forecast economic behaviors/processes in several food categories, including agricultural commodity pricing. These studies lead to the notion that anticipating agricultural commodity futures prices and behavior will reduce the uncertainty and risks associated with the existing crop of agricultural markets [24]. In the first step, machine learning models were used to approximate the price of agricultural commodities so that insurance algorithms could identify pricing and trend behaviors with accuracy. The static or non-stationary complicated behavior of input data can be captured by machine learning models [25–27]. A non-stationary characteristic of agricultural commodity

Table 1 A list of some key mathematical abbreviations used in this paper

Abbreviations	Descriptions
GA	Genetic algorithm
ELM	Extreme Learning Machine
LSTM	Long short-term memory
Auto-ARIMA	Autoregressive integrated moving average
GA-ELM	Extreme learning machine with the genetic algorithm
GA-LSTM	Long short-term memory with the genetic algorithm
RNN	Recurrent neural network
CBOT	Chicago Board of Trade
RMSE	Root-mean-square error
MAPE	Mean absolute percentage error
MAE	Mean absolute error

futures prices is that they change significantly over time as a result of several issues that we currently face or may encounter in the worldwide community. The ARIMA approach, like most methods, is more effective at modeling linear processes than nonlinear phenomena. Despite this limitation, the ARIMA may be expected to produce accurate forecasts when some agricultural products are less affected by the numerous challenges faced in the globalizing world, thus when reducing the factors that may affect the modeling results. The merits and shortcomings of the ARIMA have been discussed in many studies and from time to time comparatively discussed in many studies presented in the literature. Although it is seen that the ARIMA approach has been successfully applied in fields that appeal to a wide range of fields, it is seen that it performs effectively and successfully when it takes the stage in estimating some agricultural commodity prices, sometimes in effective combinations, and sometimes either on its own or comparatively [28–36]. Due to the nature of the data, ARIMA-based methods were seen to be relatively slow in providing the desired contribution in capturing long-term trends and short-term fluctuations in several fields of research in the following years [35]. However, it should not be overlooked that many of the recent artificial neural network models have their roots in more conventional techniques like the Auto-ARIMA [36]. With the advancement of computer and engineering technologies over time, current approaches such as LSTM, also known as soft computing methods, have recently come to the fore [23, 37]. The LSTM introduced by Hochreiter and Schmidhuber [38] has the advantage of processing long-term and non-stationary data and is widely employed in numerous disciplines of science such as finance [39], meteorological forecasts [40], energy markets [41] and transportation [42]. The GA-LSTM, which is introduced here for the long-term determination of agricultural commodity prices, is motivated by the goal of predicting financial data and outperforms the Auto-ARIMA in predicting commodity prices of some agricultural items.

The positive justifications, which are much more prominent in the historical development above, have revealed the necessity of including these two methods comparatively in the important methodological stages of this study. Like any algorithm, both conventional algorithms like the Auto-ARIMA and contemporary algorithms like the LSTM offer advantages and disadvantages. The drawback of the Auto-ARIMA is that it produces inaccurate long-term forecasts. Both the Auto-ARIMA and LSTM algorithms need a lot of training time.

The literature tells us that the ELM was introduced by Huang et al. [43, 44] to minimize these disadvantages and to combine some advantages such as regression, classification, clustering, feature learning and sparse approach. One of the most important features of the ELM is that it adds a randomness feature to the ELM due to the random assignment of input weights and biases, which improves its universality and efficiency performance over other standard models [45, 46]. Due to their advantages, the major goal of this study is to develop a new and effective model combining the GA and ELM for the prediction of agricultural commodity prices. The proposed methodology improves the ability to accurately predict agricultural commodity prices of eleven different products, including agricultural products that cause food inflation increases that affect many countries in the world, such as the USA and European countries, which we have encountered recently. In this study, the prediction capabilities of ELM are increased by automatically adjusting features such as the number of hidden nodes, layer weights, and number of lags by the GA. Thus, the superiority of the GA-ELM approach in predicting the prices of several agricultural commodities has been tested and compared with the popular and very effective Auto-ARIMA and GA-LSTM models. Furthermore, it has been found that the GA-ELM takes less time than the Auto-ARIMA and GA-LSTM models.

This study utilizes advantages of data-driven models [47–49] by using nearly two decades of price data for eleven different agricultural commodities as input, without analyzing the cause-effect relationships of variables affecting agricultural commodity prices. The ability of these models to predict agricultural commodity prices over sixty-day intervals is the main focus. While several models have achieved successful results, policymakers are concerned that the expected contributions may be limited because these forecasts are for a limited number of commodities and are short-term in nature. This study comprises eleven major agricultural commodities that have increased in price in recent years, such as wheat, corn, and soybeans. The proposed model outperforms many studies in the literature in terms of accuracy and spotting trends of sixty-day forecasts more effectively, making it a formidable rival for real-time agricultural commodity price forecasting applications. It is thought that the study will make important practical contributions to stakeholders and policymakers in the agricultural sector due to its capacity to reveal trends in both the number of products and sixty-day forecasts.

2 Material and methods

The dataset of the current models consists of two main parts as seen in Fig. 1. The first part has employed 80% of the daily data from the start date of eleven distinct agricultural commodities goods to 11.03.2022 as training data and 20% as testing data. In the second part, the models have been trained with daily data from the start date of eleven agricultural commodities to 11.03.2022. Based on this information, the prices of all agricultural products are forecasted for sixty days after 11.03.2022. The data after this date have been considered as out of sample and used in this study for the purpose of controlling the predictions.

Sustainable, smart, and precision agriculture policies are clearly crucial for countries today. Therefore, it is undoubtedly simpler to understand the significance of such forecasts for wheat, corn, sugar, soybean, rice, oat, cotton, coffee, cocoa, soybean oil, and lumber products, in the long term, both qualitatively and quantitatively, in the current time period, where prices vary widely, and inflation is a major factor in determining agricultural policies.

This section summarizes some background information needed to understand the GA-ELM, GA-LSTM hybrid methods and Auto-ARIMA approach used in this study.

2.1 Extreme learning machines (ELM)

The ELM suggested by Huang et al. [50] is a single hidden layer feed-forward neural network with a fast-learning speed and a generalization feature [44, 45, 51]. This enables the ELM to function effectively on data other than training data. More clearly, the ELM that incorporates nonlinear activation functions is successful in simulating nonlinear issues resulting from numerous external and environmental causes. The biases and weights of the hidden layer nodes are chosen at random in this method, which reduces the amount of time required for backpropagation and hyper-tuning. The weights of the output layer can be calculated analytically using the Moore–Penrose inverse [52]. Figure 2 depicts the organizational structure of the ELM model.

The following equation represents the output function of the ELM with L hidden nodes

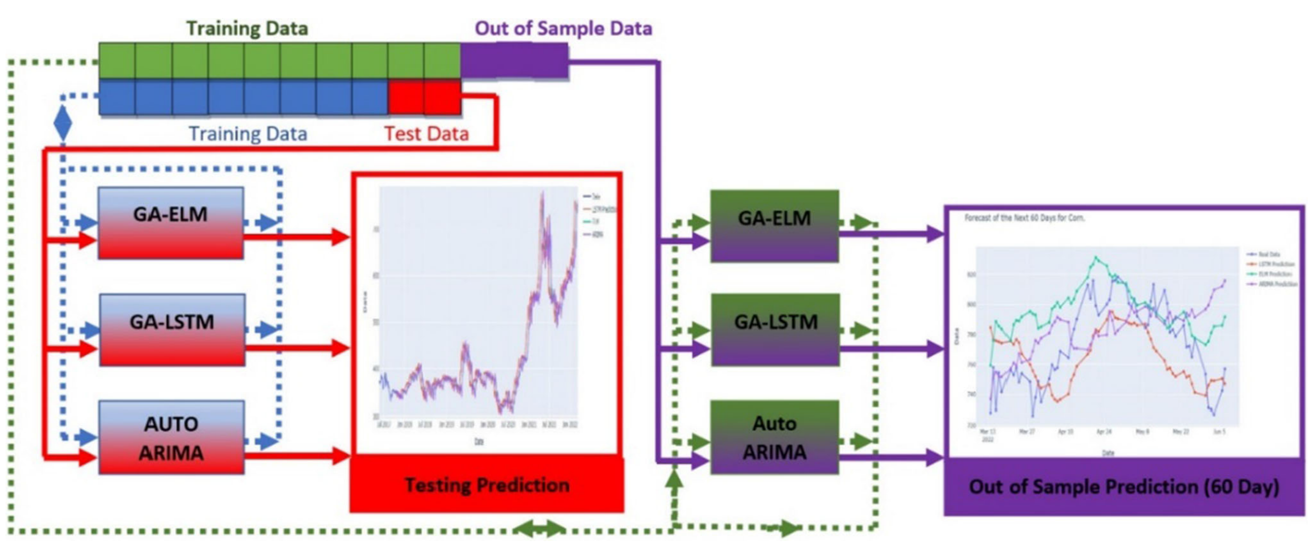

Fig. 1 Schematic diagram of the paper

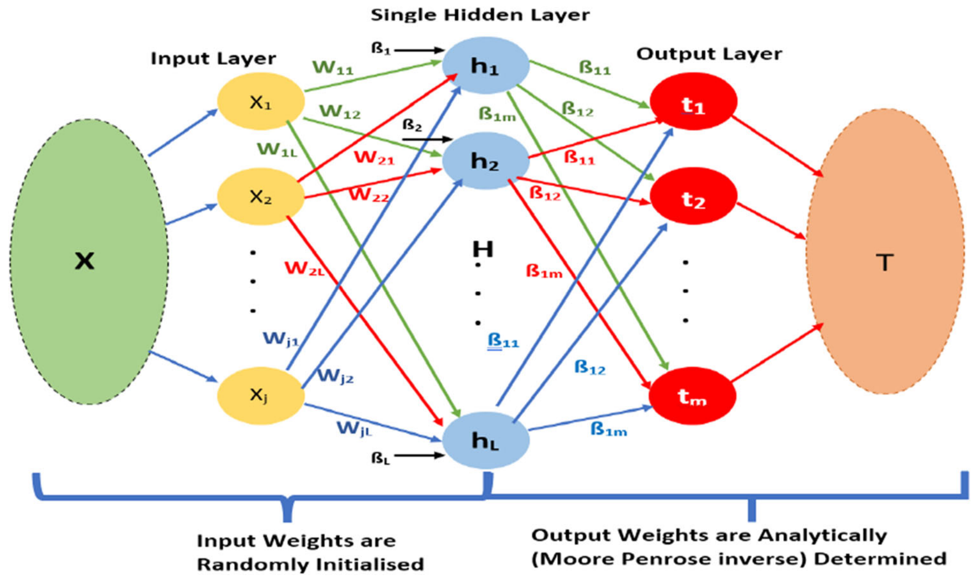

Fig. 2 The structure of the ELM algorithm

$$f_L(x) = \sum_{i=1}^{L} \beta_i h_i(x) = \sum_{i=1}^{L} \beta_i h\left(x_j * w_i + b_i\right), \quad j = 1, \ldots, N. \tag{1}$$

Here, N is the length of the training data vector. The weight of the i-th hidden layer is w_i and the bias is b_i. Here, x_j, h and β_i represent the element of the input vector j, the activation function and the weight value i between the hidden layer and the output layer, respectively.

Equation (1) can be re-expressed as

$$T = H\beta, \tag{2}$$

where

$$H = \begin{bmatrix} (x_1 * w_1 + b_1) & \ldots & (x_1 * w_L + b_L) \\ \vdots & \vdots & \vdots \\ (x_N * w_1 + b_1) & \ldots & (x_N * w_L + b_L) \end{bmatrix}, \tag{3}$$

$$\beta = \begin{bmatrix} \beta_1^T \\ \vdots \\ \beta_L^T \end{bmatrix}, \tag{4}$$

$$T = \begin{bmatrix} T_1^T \\ \vdots \\ T_N^T \end{bmatrix}. \tag{5}$$

The matrices H, $T$and $\widetilde{\beta}$ represent the hidden layer output matrix, the training data vector and the least-squares solution of the linear system $H\beta = T$, respectively. The matrices $H^{\dagger}$ and T stand for the Moore–Penrose generalized inverse of H and the training data vector, respectively. Then,

$$\tilde{\beta} = H^{\dagger} T. \tag{6}$$

To make predictions with testing data or out of sample data (O), it can be written as:

$$F(O) = \widetilde{\beta} H(O). \tag{7}$$

When it comes to time series forecasting, the ELM algorithm can be an effective tool [53]. Statistical time series forecasting models (ARIMA, for example) are algorithms designed for stationary time series. These algorithms frequently fail to detect seasonality in time data. Because they contain activation functions, ANN models have universal approximation capabilities. They can overcome the difficulties of capturing environmental and external impacts, i.e. nonlinear circumstances, in this manner. Aside from this benefit, ELMs have extraordinarily quick learning rates, good generalization performance, and require little human involvement [46]. The ELM algorithm combines the benefits of traditional back propagation neural network algorithms with the elimination of some of the disadvantages of ANN, such as overfitting and slow learning speed.

The ELM method may perform less accurately than ideal due to stochasticity. The amount of lag, type of activation function, number of hidden nodes, and other hyperparameters must be properly configured in order to prevent this issue. Meta-heuristic optimization algorithms like GAs and Bayesian Optimization can be used for this tuning.

2.1.1 Genetic algorithm-based extreme learning machine

Because the weights and bias of the input layer are created at random in the ELM, the system may exhibit unsteady behavior. Because the weights and biases of the ELM input layer are randomly generated, there may be too many nodes in the hidden layer, causing overfitting during the training process. The GA can choose to optimize the weights and bias of the input layer of the ELM, thus improving the stability of the model and the prediction accuracy.

The GA offered by Holland as a solution to the optimization problems of complicated nonlinear systems [54] seeks the best solution in an adaptive manner. It may automatically collect information about the search area. The citation above contains more information about the procedure.

In this study, the GA is used to optimize the number of input lags, the type of activation function, and the number of hidden nodes parameters, in addition to the input weights and biases in the ELM algorithm. The GA-ELM technique for commodity price forecasting is depicted in Fig. 3. The data preparation and selection of hyperparameters to be optimized in the methodological development of the GA-ELM are shown in Table 2 and Fig. 3. The implementation of optimization and iterative model parameter generation are shown in Fig. 3 together with the random generation of learning parameters. The literature [55–59] contains structural information on the hyperparameters utilized in this study, including the lag size, number of hidden nodes, input weights/biases, and activation function of the ELM. It should be noted that the selection strategy used in this study was roulette and the GA was constructed using two-point crossover and Gaussian mutation as suggested by Immanuel and Chakraborty [60].

Table 2 Selection of hyperparameters to be optimized

Variable	Type	Lower bound	Upper bound
Number of input lags	Integer	1	200
Number of hidden nodes	Integer	1	500
Input weights	Double	− 5	5
Input biases	Double	− 5	5
Type of activation function	Integer	1	4

2.2 Long short-term memory (LSTM)

The LSTM is an artificial recurrent neural network (RNN) used in deep learning [61]. The LSTM was presented by Hochreiter and Schmidhuber to access constant error carouse units and problem of gradient disappearance [38]. The LSTM is a deep learning model that has recently been widely used in time series forecasting and is specifically designed to solve problems such as gradient bursting that arise when working with long sequence data [62]. The

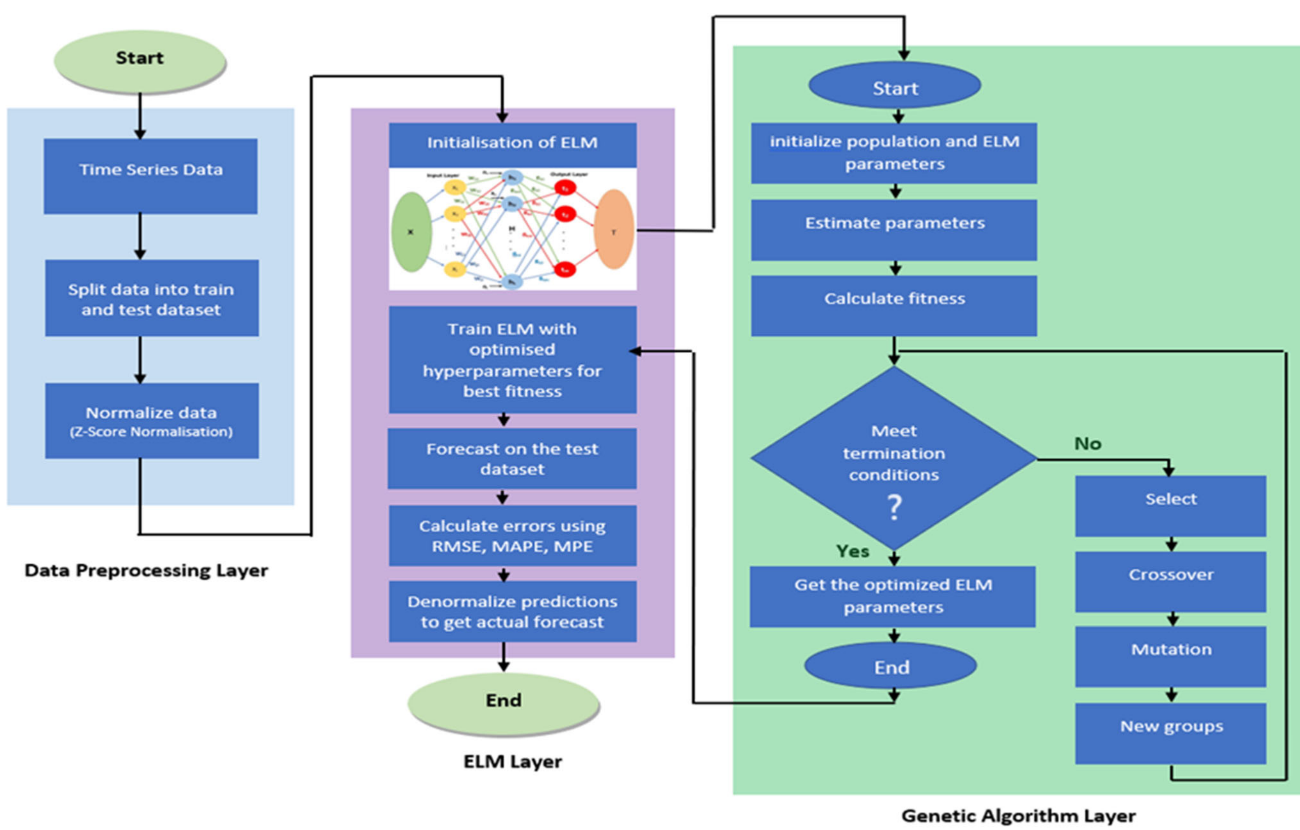

Fig. 3 The GA-ELM architecture

main concept of LSTM is based on cell state and gate structures. It can be seen that these structures allow the model to learn long-term dependencies more effectively and produce more consistent results [63]. In the first version of the LSTM block, the cells include input and output gates. The LSTM has feedback connections as opposed to conventional feed-forward neural networks. The fact that it can analyze data arrays in addition to instant data is a significant advantage. When dealing with estimate of nonlinear and time series problems, the LSTM model with memory capabilities can provide significant advantages [64]. To remember prior information, an LSTM memory cell structure is added, and three gate structures are introduced to manage the transfer of historical knowledge: input gate, output gate, and forget gate. The cell stores values in variable-length time periods, and these three gates control and regulate the flow of information entering and leaving the cell. The structure of the LSTM algorithm at time step t is depicted in Fig. 4.

Assuming the network input is $x_1, x_2, \ldots, x_t$ and the hidden-layer state is $h_1, h_2, \ldots, h_t$, the computations of each unit and gate at time t are presented as follows.

The LSTM cell has three gates that play an important role in controlling the flow of information through the neural network. Three gates on the LSTM cell are crucial in regulating how information moves through the neural network. Input, output, and forget gates are the names of these gates. The input gate (i_t), which controls how much input data is current state stored in the unit state, is the first of them and

$$i_t = \sigma(w_i.[h_{t-1}, x_t] + b_i), \tag{8}$$

where h_{t-1}, w_i and b_i represent the output values of the previous cell, the weight matrices of the input gate and the bias of the input gate, respectively.

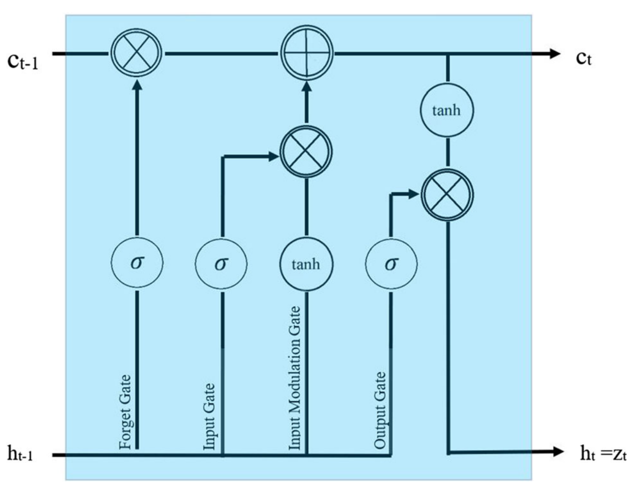

Fig. 4 The structure of the LSTM algorithm

The second of these is the forget gate (f_t) and is calculated as follows:

$$f_t = \sigma(w_f.[h_{t-1}, x_t] + b_f), \tag{9}$$

where w_f and b_f indicate the weight matrices of the forget gate and the bias of the forget gate, respectively.

The most important step in creating an LSTM network is to determine how much information is retained in the current cell state of c_t, from the state of c_{t-1} from the previous moment. In other words, it is the determination of the information that is not necessary and will be removed from the cell:

$$c_t = f_t \circ c_{t-1} + i_t \circ \tilde{c}_t, \tag{10}$$

$$\tilde{c}_t = \tanh(w_c.[h_{t-1}, x_t] + b_c), \tag{11}$$

where c_t and $\tilde{c}_t$ stand for the current input and current moment unit state, respectively.

The third gate (o_t) is the output gate and is calculated as:

$$o_t = \sigma(w_o.[h_{t-1}, x_t] + b_o), \tag{12}$$

$$h_t = o_t \circ \tanh(c_t), \tag{13}$$

where w_o and b_o indicate the weight matrices of the output gate and the bias of the output gate, respectively.

The notations $\sigma(.)$ and $\tanh(.)$ indicate transfer functions;. and $\circ$ represent the inner product and element-wise multiplication, respectively.

2.2.1 Genetic algorithm-based LSTM (GA-LSTM)

In order to increase the performance of deep neural networks, hyperparameters are usually determined by trial-and-error method. This method is time consuming as well as data dependent. In this study, the GA has been used to optimize the LSTM architectural parameters for each of the eleven products. Optimal lag size and number of units parameters of LSTM architecture are optimized by genetic algorithm. The architecture used in the study is shown in Fig. 5.

2.3 Autoregressive integrated moving average (ARIMA)

ARIMA models are stable which implies stationarity. ARIMA model can be applied to non-stationary series after differencing them up to stationary. One of the most important features of this model is that it is based on a linear function between past observations and random errors. The ARIMA model is a combination of AR and MA models. p and q describe the order of AR and MA models, respectively, while d is the degree of difference [65]. Then,

$$\delta_t = \sum_{i=1}^{p} \phi_i \delta_{t-i} - \sum_{j=0}^{q} \varphi_j \varepsilon_{t-j}. \tag{14}$$

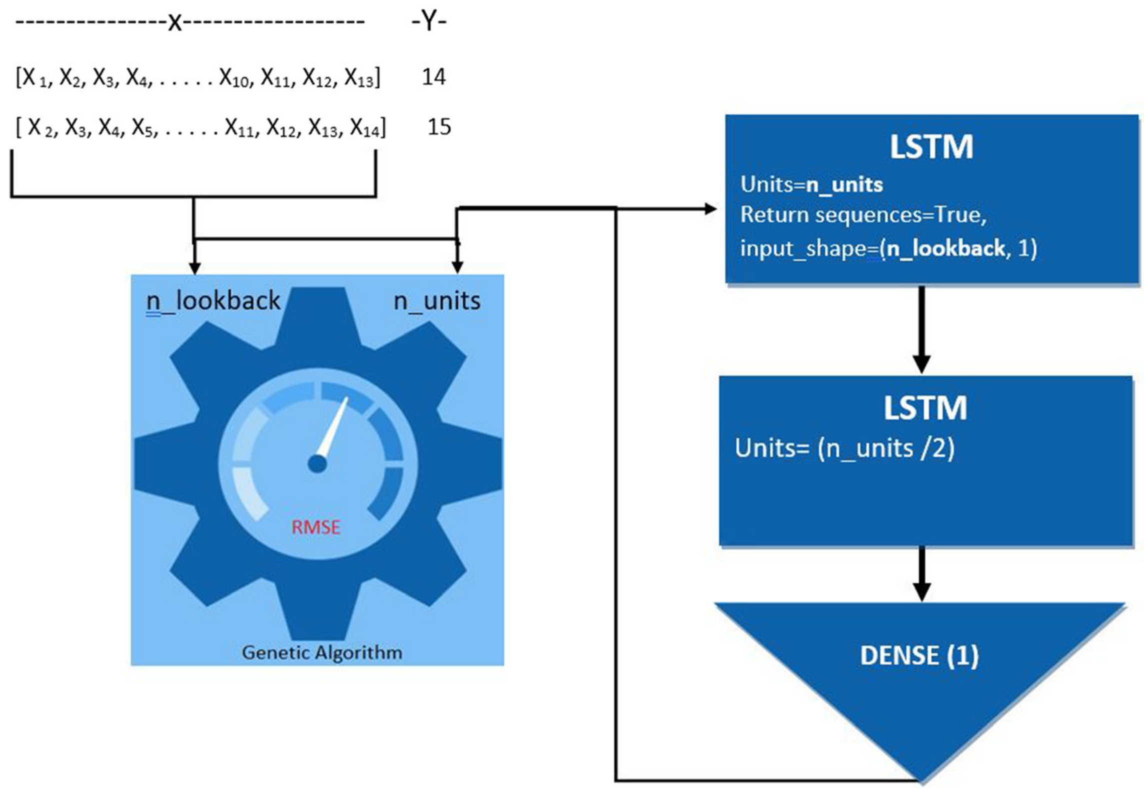

Fig. 5 The architecture of the GA-LSTM algorithm

Here, δ_t denotes the actual value of the time series at time t. δ_{t-i} and ε_{t-j} represent different observation values and error terms, respectively. The coefficients $\phi_i(i = 1, 2, ..., p)$ and $\varphi_j(j = 1, 2, \ldots, q)$ are the parameters of the AR and MA models, respectively. Details of the ARIMA modeling procedure, for instance, can be found in the work of Box and Jenkins [65]. Also note, by the way, that the Auto ARIMA method returns the best ARIMA pattern based on the AIC, AICc, or BIC value. The method performs a search on the possible model within the provided ordering constraints and finds the ARIMA model with the best parameters.

3 Experiments and analysis

3.1 Data processing and analysis

This paper has analyzed eleven different data groupings, including daily data for wheat, corn, sugar, soybeans, rice, oat, cotton, coffee, cocoa, soybean oil, and lumber agricultural items in Table 3. The Chicago Board of Trade (CBOT) provides data on wheat, corn, soybean, rice, oat, and soybean oil agricultural products, while the Immigration and Customs Enforcement (ICE-US) provides data on sugar, cotton, coffee, and cocoa agricultural products for daily sales prices. All agricultural commodity unit prices are expressed in US dollars. Furthermore, there are gaps in the data utilized on some days owing to holidays or technical issues, which are not included in this study.

The use of data from reputable sources such as CBOT and ICE-US enhances the reliability of this paper. In terms of data pre-processing, the dataset has carefully been curated and cleaned to ensure accuracy and consistency. However, it is important to recognize that some limitations and biases may remain despite rigorous pre-processing efforts. Notably, the study addresses potential gaps in data due to holidays or technical issues on certain days, and such cases are excluded from the analysis to maintain data integrity. However, it is imperative to recognize that inherent biases or limitations in the original data sources may affect the results of the study. This article provides a comprehensive understanding of the quality and limitations of the dataset, as well as transparency about the source of the data, pre-processing steps, and potential biases.

Based on literature reviews [66, 67], prediction periods used in machine learning models are generally categorized as short, medium and long term. However, the limitation that the meanings of these terms may vary from context to context should be noted. In this study, we classify sixty-day forecast periods as long-term, and forecasts shorter than two weeks as short-term. A medium-term time period was

Table 3 Descriptive statistics on agricultural commodity products

	Time period	Number of record	Mean	St. Dev	Min	Max
Wheat	01.01.2000–29.07.2022	5899	523.70	189.98	233.5	1425.25
Corn	03.01.2000–29.07.2022	5732	393.69	160.71	174.75	831.25
Sugar	03.01.2000–29.07.2022	5730	14.26	5.81	4.65	35.31
Soybean	03.01.2000–29.07.2022	5846	967.26	334.47	418	1769
Rice	19.01.2007–29.07.2022	3904	13.19	2.39	9.12	24.46
Oat	15.07.2008–29.07.2022	3481	318.91	114.31	158.50	807
Cotton	14.10.2009–29.07.2022	3288	83.09	24.92	48.85	213.84
Coffee	03.01.2000–29.07.2022	5721	126.81	50.20	41.50	304.90
Cocoa	03.01.2000–29.07.2022	5677	2208.14	670.01	674	3774
Soybean oil	03.01.2000–29.07.2022	5745	35.04	14.09	14.38	90.60
Lumber	03.01.2000–29.07.2022	5686	351.92	193.65	138.10	1686

not taken into account in this study. However, it is imperative to recognize that time constraints may affect the results of the study.

3.2 Performance metrics

Our forecasting performance relies on several key metrics: root-mean-square error (RMSE), mean absolute percentage error (MAPE), and mean absolute error (MAE) [68]. The RMSE is calculated as the square root of the average squared differences between the actual values (yi) and the forecasted values $(\hat{y}_i)$, summed over all observations (N). Specifically, the RMSE, MAPE and MAE are given by:

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}, \tag{15}$$

where y_i, $\hat{y}_i$ and N stand for the actual data, the forecasted data, and the overall number of days that the forecasting was done for. Also

$$\mathrm{MAPE} = \frac{100}{N}\sum_{i=1}^{N}\left|\frac{y_i - \hat{y}_i}{y_i}\right|, \tag{16}$$

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|. \tag{17}$$

4 Application of the models

The data are separated into two categories: training data and test data. 80% of the data according to the start date has been employed in the training time of the GA-LSTM and GA-ELM models proposed with the Auto-ARIMA. The remaining 20% has been used for testing in all models. Future data for eleven different agricultural product prices have been predicted using three different models as shown in Figure 6.

Forecast performance is measured by using root-mean-square error (RMSE), mean absolute error (MAE) and mean absolute percentage error (MAPE) as seen in Table 4.

4.1 Approaches for forecasting agricultural commodity prices over the next sixty days

This section determines the data in agricultural commodity prices before to 12.03.2022, which is the second key theme of the study, and presents forecasts for the following sixty days. The prediction results, which can be regarded very long, have been determined on a daily basis for all items from the start date of 13.03.2022 to the finish date of 05.06.2022. Because weekend data are not available between these periods, it has been examined over a sixty days period. Furthermore, projections are not produced on the days when real data are unavailable.

In Fig. 7, it can be observed that the GA-ELM model for wheat, which is one of the products most affected by the food crisis caused by global conflicts, generates very successful behavioral predictions when compared to the GA-LSTM and Auto-ARIMA models. Although known to be quite successful overall, the Auto-ARIMA lagged undeniably behind the GA-ELM. It has been found that GA-LSTM could not consistently approach real data, both in terms of quality and quantity, depending on the structure of the data.

In comparison to the GA-LSTM in Fig. 8, it can be shown that the GA-ELM and Auto-ARIMA models for corn, which is included in cereal products, both generate very accurate quantitative predictions. The GA-ELM model is shown to come to the fore both qualitatively and quantitatively according to the Auto-ARIMA, especially when looking at the long-term forecasts made after May 22. It is worth noting that the GA-ELM successfully

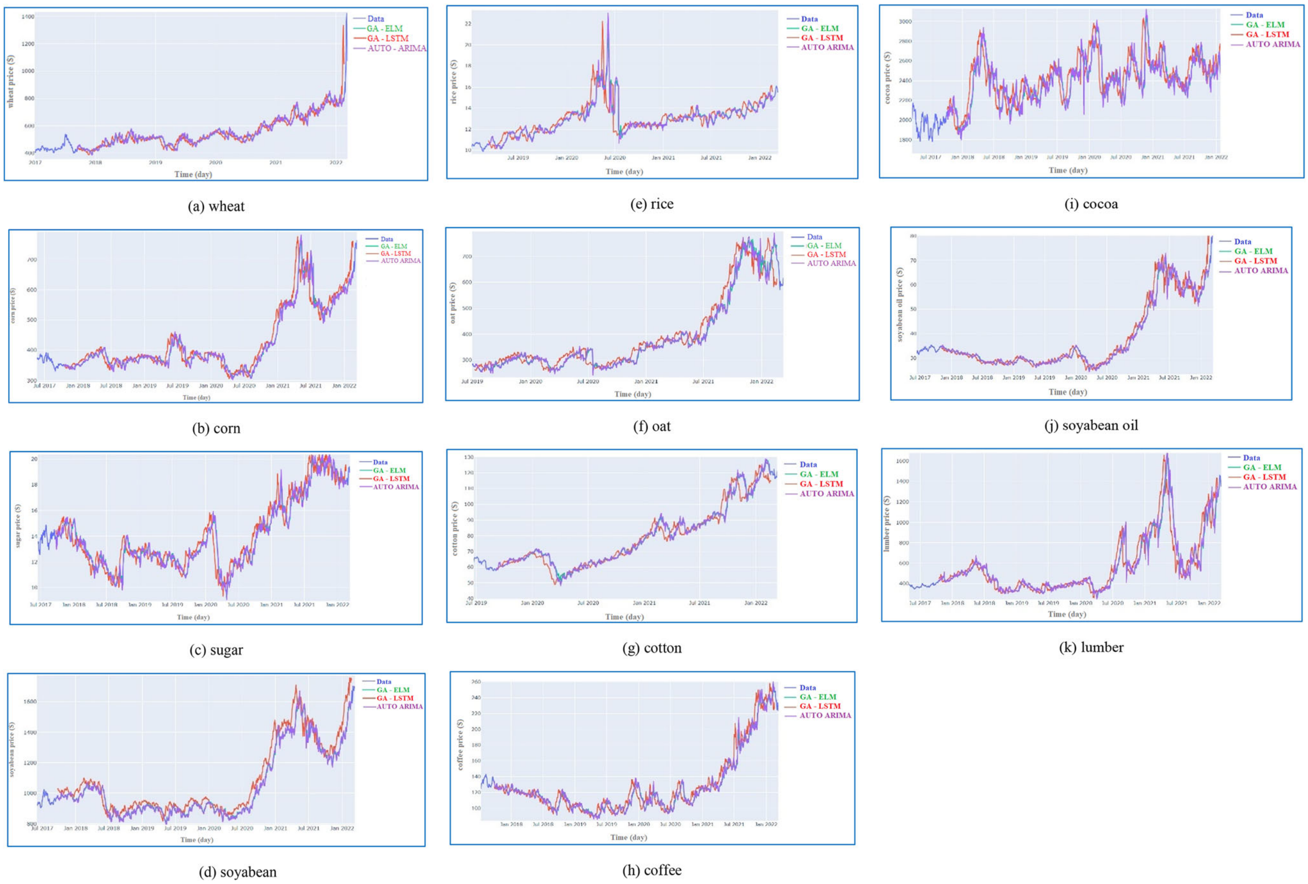

Fig. 6 Forecasting results of the GA-ELM, GA-LSTM and Auto-ARIMA models

Table 4 Forecasting results for agricultural commodity futures

	GA-ELM			GA-LSTM			Auto-ARIMA		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE
Wheat	**10.27**	**7.82**	**1.39**	43.52	25.45	4.42	10.91	8.36	1.48
Corn	**8.36**	**5.15**	**1.14**	32.38	20.75	4.58	8.96	5.49	1.22
Sugar	**0.25**	**0.19**	**1.37**	0.90	0.69	5.5	0.26	0.20	1.46
Soybean	**14.18**	**9.99**	**0.93**	47.04	35.12	3.24	16.47	11.31	1.05
Rice	**0.30**	**0.15**	**1.09**	1.07	0.60	1.10	0.31	0.15	1.14
Oat	**11.82**	**7.32**	**1.72**	38.28	27.04	6.63	11.91	7.41	1.81
Cotton	**1.53**	**1.05**	**1.37**	5.17	3.98	4.90	1.55	1.15	1.43
Coffee	**2.94**	**2.11**	**1.64**	9.89	7.12	5.64	3.10	2.23	1.73
Cocoa	**50.60**	**36.78**	**1.52**	161.59	122.22	5.10	54.07	39.46	1.63
Soybean oil	**0.71**	**0.46**	**1.11**	2.96	1.86	4.37	0.75	0.49	1.19
Lumber	**28.17**	**16.67**	**2.62**	112.62	73.10	11.10	30.09	16.46	2.79

captures the behavioral tendencies of real data, even when global crises directly boost agricultural commodity prices. Furthermore, the GA-LSTM model is slower than the other two models in capturing long-term forecasts.

Considering the changes in sugar prices in the last twenty-two years, the change in prices is low since it has a homogeneous structure compared to many other agricultural commodities. Therefore, depending on the nature of the data, all three models appear to be successful in the first five-day forecasts (short-term) in Fig. 9. Despite the fact that prices in real data exhibit relatively very little fluctuations beyond five days, the GA-ELM model is more sensitive than the other two models in capturing these trends.

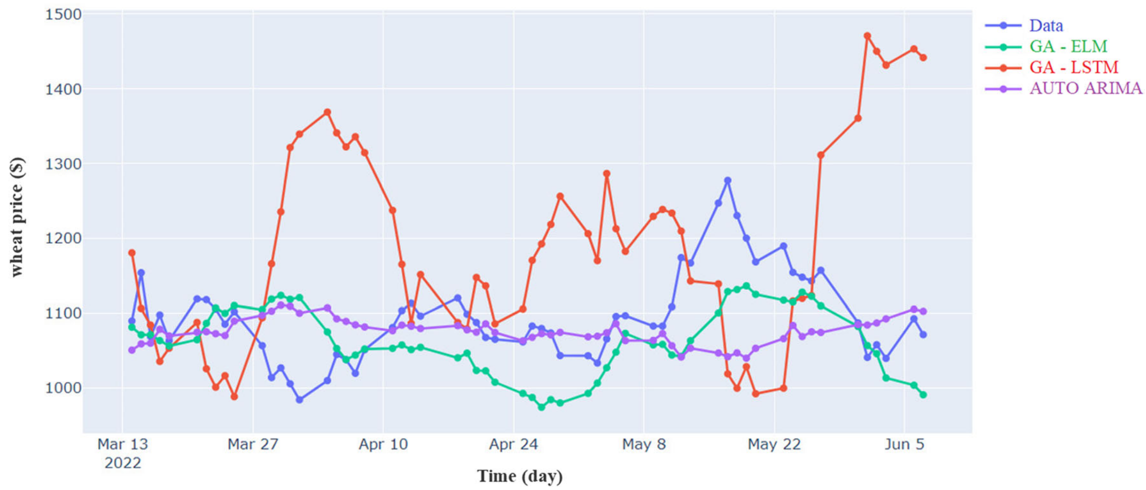

Fig. 7 Forecasts of the models for wheat for the next 60 days

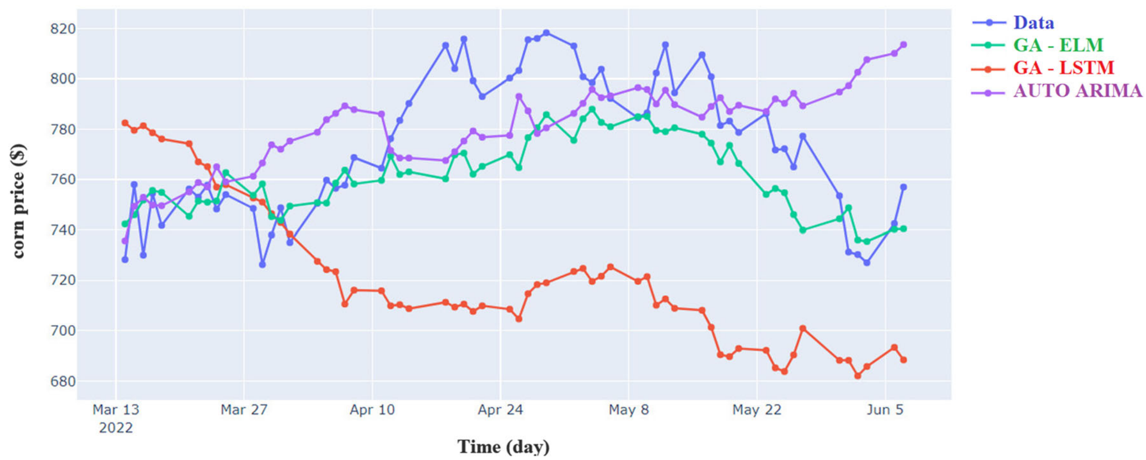

Fig. 8 Forecasts of the models for corn for the next 60 days

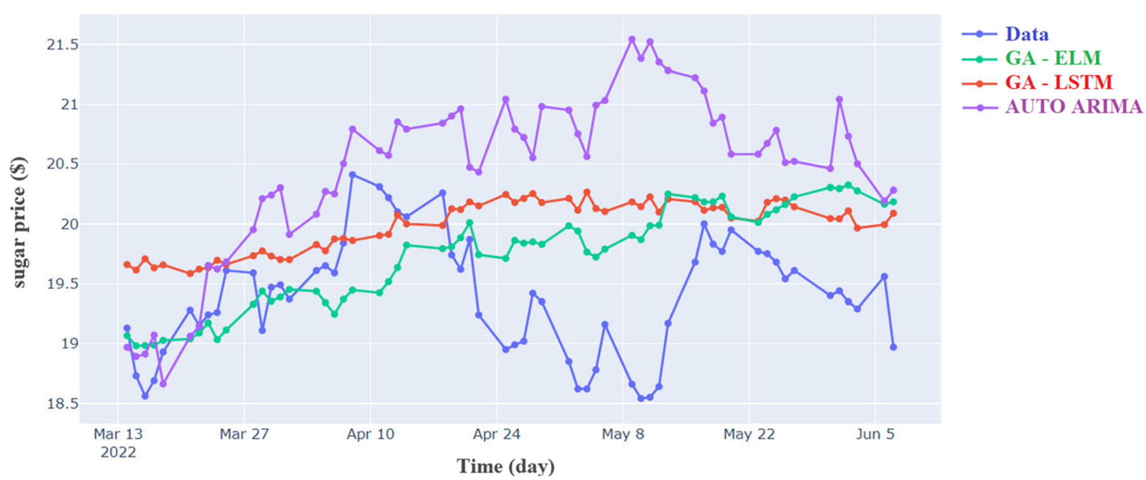

Fig. 9 Forecasts of the models for sugar for the next 60 days

When we look at the price fluctuations of soybean prices over the previous twenty-two years, we see that the price range is fairly wide since it has a heterogeneous structure compared to other agricultural commodities. The GA-ELM model is quantitatively successful in the long-term forecasts shown in Fig. 10. Furthermore, when the 5846-day data are examined, it is apparent that GA-ELM caught this upward trend in the long run on 09.06.2022, when soybean prices were at their highest. Although the Auto-ARIMA has shown an upward trend in long-term forecasts, it has not shown the predicted performance quantitatively. Contrary to the Auto-ARIMA, the GA-LSTM was expected to have a downward trend in soybean prices in the long run.

Since rice has a uniform structure in comparison to other agricultural commodities, price ranges are rather minimal when looking at price movements over the past twenty-two years. The long-term forecasts in Fig. 11 demonstrate the effectiveness of the GA-ELM model. In addition, it is seen that the GA-ELM and Auto-ARIMA models are faster than the GA-LSTM in catching up and down trends in prices.

Russia is one of the top producers of oat in the world. In light of the recent disputes between Russia and Ukraine, the price of oat appears to have peaked on April 12, 2022, when looking back over the previous fifteen years. It is crucial that the GA-ELM model being applied captures the upward trend shown in Fig. 12 and takes values that are very close to the largest value it has ever taken on this date. The abrupt uptrends and downtrends that happen in unforeseen situations and in capturing them, like with other agricultural commodities, make the GA-ELM stand out visibly. Although it is a little slow, GA-LSTM has been seen to play a significant role in capturing the changing trends of oat prices. Although the Auto-ARIMA is not as good as expected at capturing the ups and downs in real data, depending on the structure of the data, it has been noted that it can quantitatively capture values close to real data in long-term forecasts.

The GA-ELM model outperforms the rival models in terms of quality and quantity during the short term (first fifteen days), despite the anticipated values for cotton being less accurate than those for other agricultural products. According to Fig. 13, the GA-ELM model performs better than the other two models in capturing real values, and it is seen that it can slow down even though it maintains its superiority compared to other models even in projections longer than fifteen days.

In long-term forecasts covering sixty days, the GA-ELM model outperforms the other two models in coffee prices, which are less affected by global disputes than other agricultural products. Figure 14 illustrates how the GA-LSTM model has been slow to detect long-term up and down trends after May 8 even if it has been qualitatively good in forecasting up to that point. Additionally, after May 8th, the Auto-ARIMA has been seen to outperform the GA-LSTM model in terms of quality and quantity in long-term projections.

The GA-ELM model is extremely successful in capturing both the short-term and long-term ranges of variation for the cocoa product in Fig. 15. Despite the fact that the GA-LSTM model can be seen to be getting close to the trends of real data in some time intervals, it still behind the GA-ELM overall. As can be seen in Fig. 15, the Auto-ARIMA has not been successful in predicting cocoa prices. The fact that the range of changes in cocoa prices over time is so wide is one of the most significant elements generating this situation.

Soybean oil is one of the products most affected by the recent global problems in the world. As can be seen in Fig. 16, soybean oil reached its peak value on April 28, 2022. The sixty-day projections (long-term) covering this

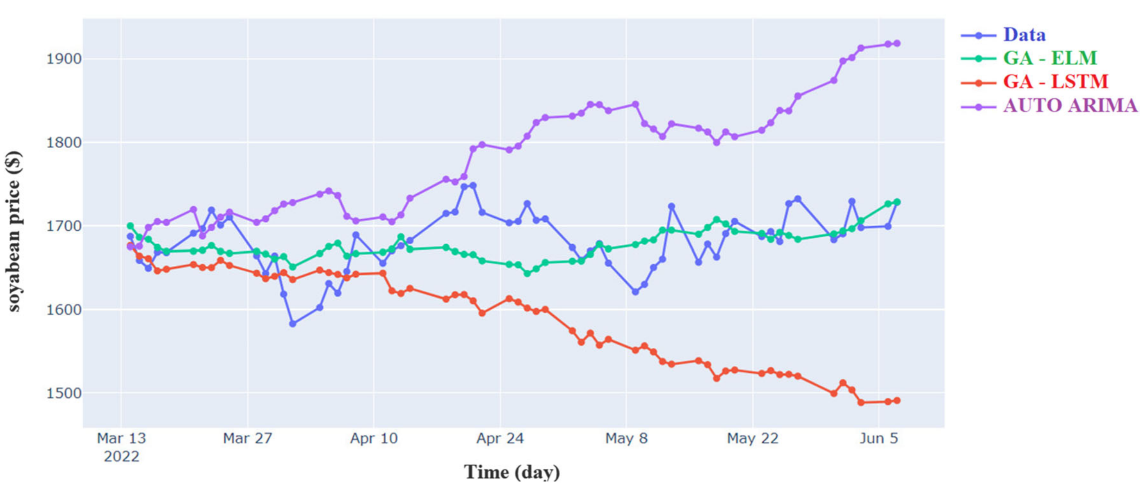

Fig. 10 Forecasts of the models for soybean for the next 60 days

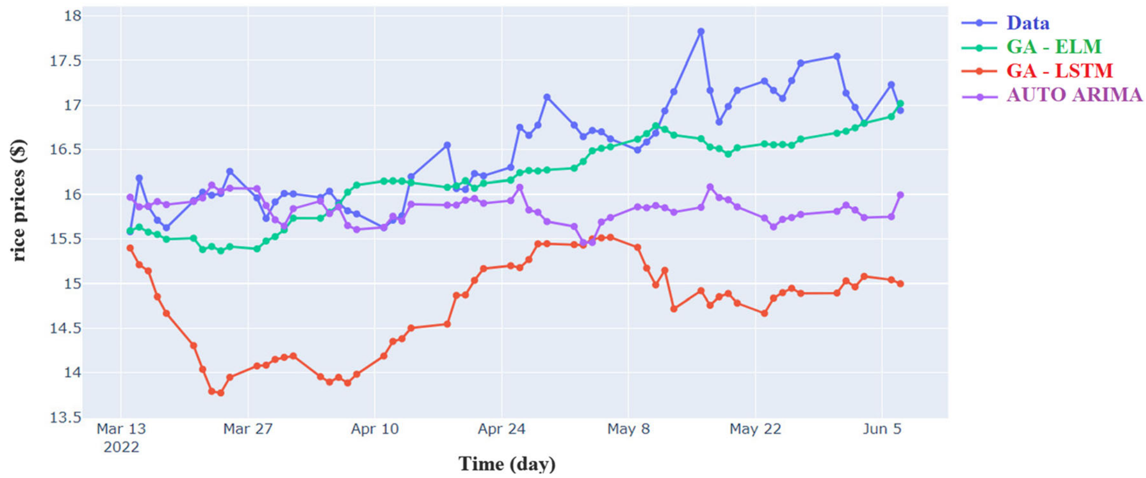

Fig. 11 Forecasts of the models for rice for the next 60 days

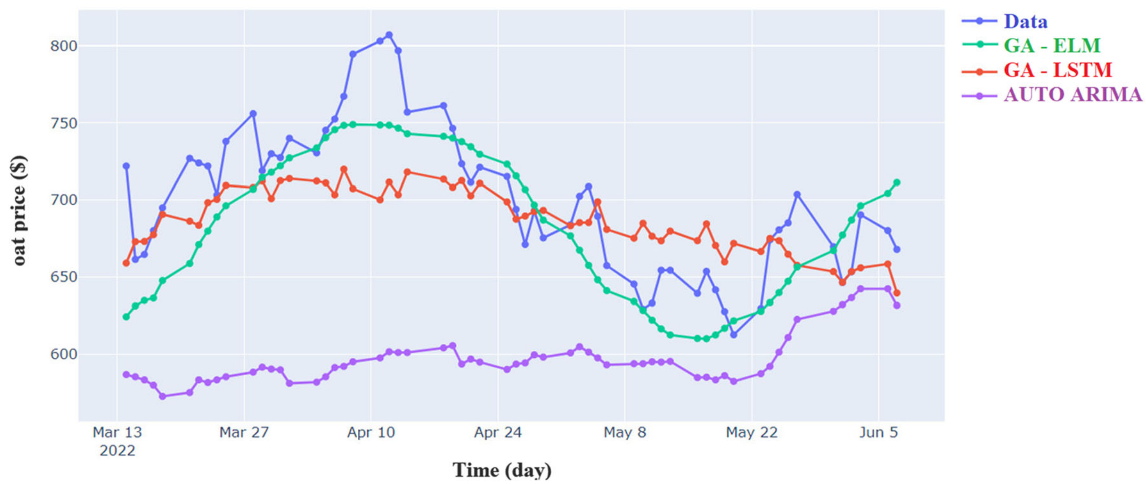

Fig. 12 Forecasts of the models for oat for the next 60 days

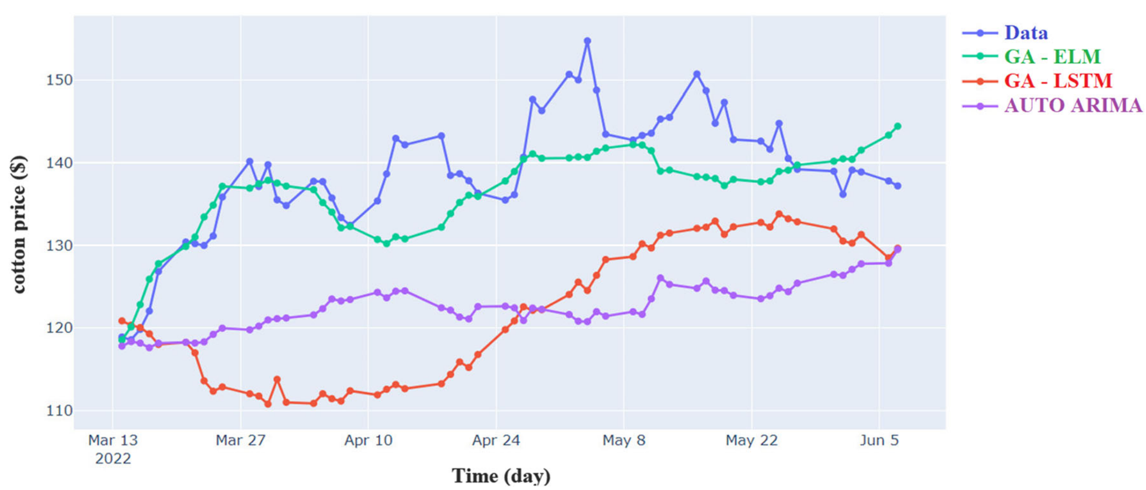

Fig. 13 Forecasts of the models for cotton for the next 60 days

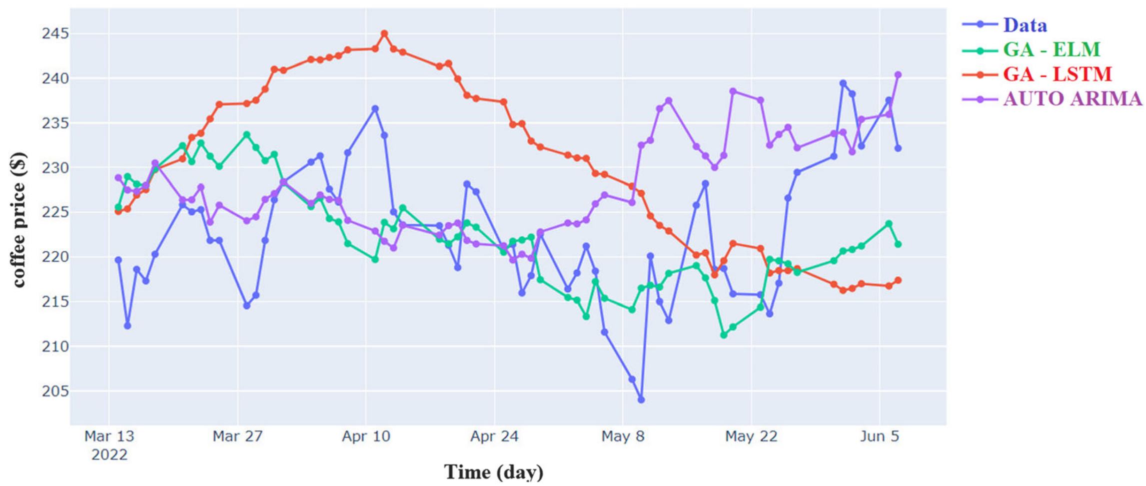

Fig. 14 Forecasts of the models for coffee for the next 60 days

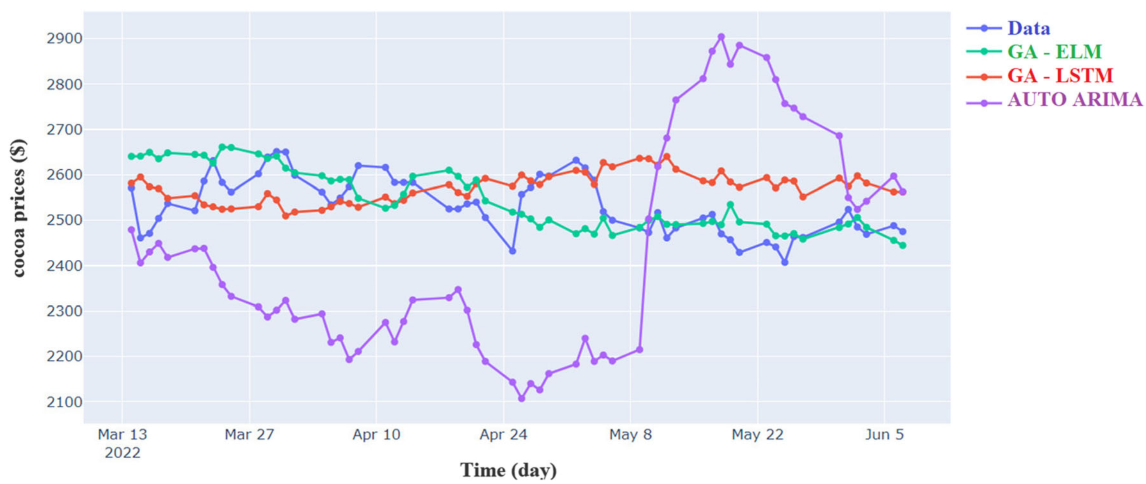

Fig. 15 Forecasts of the models for cocoa for the next 60 days

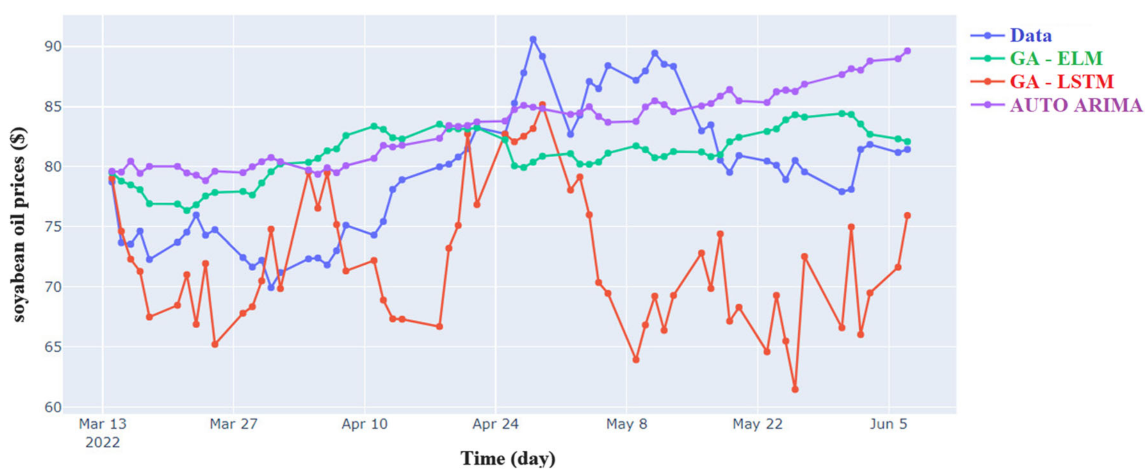

Fig. 16 Forecasts of the models for soybean oil for the next 60 days

date show that GA-LSTM has particularly well captured this increasing trend. But when looking at a longer time period, it is clear that the GA-ELM model excels at identifying these trends. The Auto-ARIMA is seen to be successful in forecasting till May 22 while being very slow to identify rapid changes.

The GA-ELM model for lumber prices is quite successful in both the long and short terms, as can be observed in Fig. 17. Additionally, even though the GA-LSTM is effective in the near term, it is noticed that it was slow to reach the expected performance between April 21 and May 11. The GA-LSTM seems to show a downward trend in performance after May 11. Compared to both competing models, the Auto-ARIMA has been slow to achieve expected performance, both qualitatively and numerically, as shown in Fig. 17.

As demonstrated in Table 5, the GA-ELM model performs better and has a higher sensitivity when compared to the Auto-ARIMA and GA-LSTM models for all agricultural commodity items when the long-term forecast results covering sixty days are analyzed. The Auto-ARIMA appears to be the second efficient model after the GA-ELM for wheat, corn, rice, cotton, coffee and soybean oil products. Apart from these products, the GA-LSTM model stands out as the second model after GA-ELM in the agricultural products (sugar, soybeans, oat, cocoa and lumber). Although the Auto-ARIMA and GA-LSTM models are very slow in reaching the accurate results achieved by the GA-ELM due to some ranges and data types, the qualitative and quantitative predictions produced by these two models should not be perceived as useless.

Figure 18 presents a violin plot of the RMSE values of the three methods for sixty-day forecasts of sugar, rice,

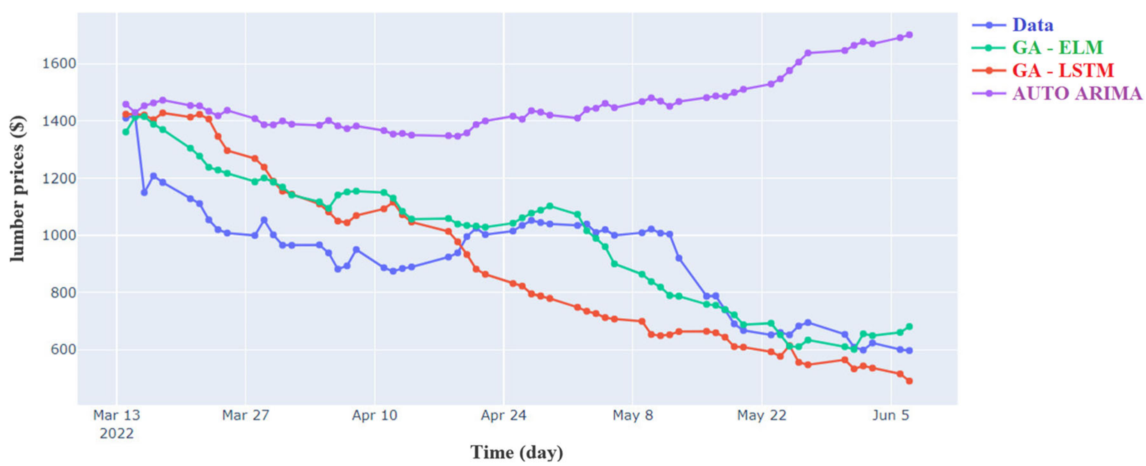

Fig. 17 Forecasts of the models for lumber for the next 60 days

Table 5 The GA-ELM, GA-LSTM and ARIMA test results for 60 days forecasts of agricultural commodity prices

	GA-ELM			GA-LSTM			Auto-ARIMA		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE
Wheat	**65.94**	**53.43**	**5.13**	191.84	152.12	13.41	74.30	53.24	4.74
Corn	**21.98**	**17.73**	**2.26**	69.82	61.99	7.90	29.10	22.60	2.95
Sugar	**0.68**	**0.56**	**2.95**	0.80	0.66	3.43	1.32	1.09	5.67
Soybean	**37.43**	**29.91**	**1.73**	117.34	97.42	5.68	120.70	102.60	6.12
Rice	**0.46**	**0.38**	**2.33**	1.80	1.72	10.51	0.88	0.68	4.05
Oat	**33.86**	**27.39**	**3.98**	37.44	28.94	4.08	114.86	103.190	14.36
Cotton	**5.56**	**4.23**	**2.79**	18.92	17.07	12.47	17.53	16.20	11.46
Coffee	**8.52**	**6.93**	**2.98**	13.86	12.57	5.71	10.35	7.71	3.53
Cocoa	**70.07**	**53.65**	**2.16**	91.58	77.46	3.17	295.93	267.67	10.54
Soybean oil	**5.49**	**4.72**	**5.98**	10.38	8.43	10.34	5.64	5.02	6.48
Lumber	**138.54**	**111.09**	**19.40**	208.30	183.87	19.40	592.30	533.78	66.54

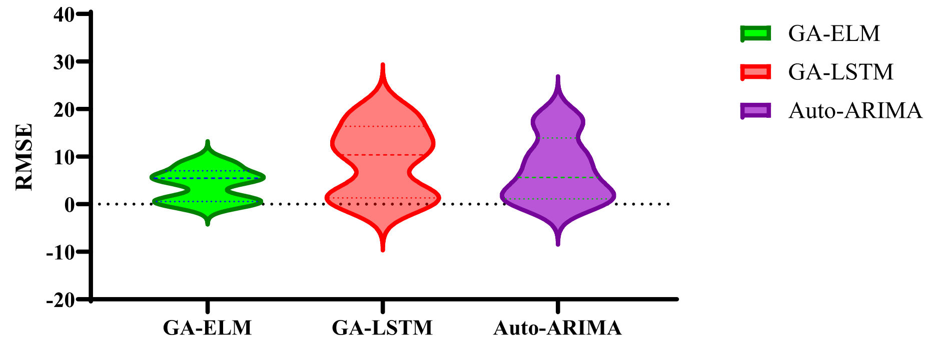

Fig. 18 Violin plot for the three proposed methods (RMSE < 20).

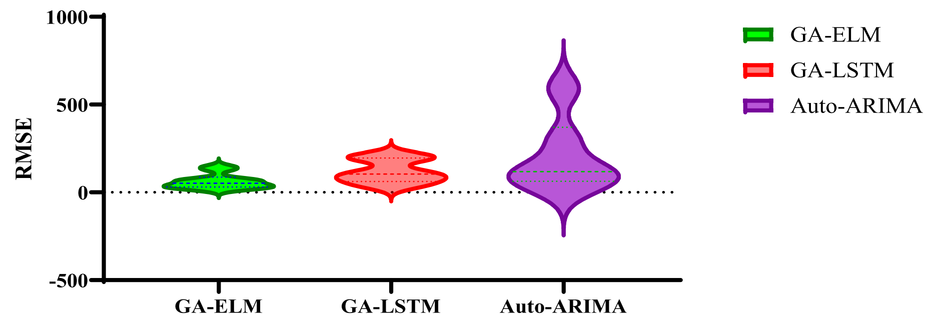

Fig. 19 Violin plot for the three proposed methods (RMSE ≥ 20).

cotton, coffee, and soybean oil prices (RMSE < 20). It is clear that the GA-ELM model provides the best distribution and fit of the prediction. RMSE values are presented comparatively for the three models. Upon close examination of Fig. 18, the Auto-ARIMA emerges as the second most effective model for capturing changes over the sixty days period following the GA-ELM.

Figure 19 presents a violin plot of the RMSE values of the three methods for sixty-day forecasts of wheat, corn, soybeans, oat, cocoa, and lumber prices (RMSE ≥ 20). It is clear that the GA-ELM model provides the best distribution and fit of the predicted RMSE values compared to the competing models (GA-LSTM and Auto-ARIMA). Upon closer inspection of Fig. 19, the GA-LSTM emerges as the second most effective model in capturing changes over the sixty-day period, following the Auto-ARIMA.

Determining acceptable limits of RMSE can be difficult when working on a variable and dynamic data set such as agricultural commodity prices. In this study, the constraint criterion for the RMSE value is defined as 20 in order to examine the effectiveness of three different methods in more detail and to categorize agricultural commodity products. As seen in Figs. 17 and 18, it can be seen that the GA-ELM model is the model that most effectively captures changes in two different agricultural commodity categories.

The RMSE values of the GA-ELM model for wheat, corn, soybeans, oat, cocoa and lumber compared to other agricultural products can be considered a successful level, considering that the forecast period is sixty days. Also note that when working on a variable and dynamic data set such as agricultural commodity prices, it can be difficult to determine acceptable limits of RMSE. High RMSE values may be caused by factors such as the complexity of the model, the characteristics of the data set or the optimization algorithms used [69]. At the same time, the fact that the RMSE values of the GA-ELM model for eleven different agricultural products are lower than the GA-LSTM and ARIMA models shows that the GA-ELM model has a superior performance. It should be noted that in addition to the quantitative prediction of agricultural commodity prices, the proposed models also successfully capture qualitative fluctuations, as shown in Figs. 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 and 17.

5 Results and discussion

Although this study focuses on agricultural product price forecasts, it should not be ignored that these results may be slightly below the desired level of success from time to time, seasonal and political considerations, as well as model-based elements, can be effective in this. At least intuitively, it can be stated that the findings achieved by eliminating the influences of periodic and political factors, as well as the effects of long-term data, can be carried to a considerably higher degree of success.

This research has presented deep learning techniques to produce predictive models of the long-term trends of agricultural commodity prices, including those of products involved in the current food crisis. Despite the fact that a variety of machine learning algorithms have been utilized in the literature to estimate agricultural commodity prices, to the best of the authors' knowledge, sixty-day forecasts have not yet been taken into account. One of the most important reasons for this situation is that the success levels of many models used decrease as the prediction time increases. As a result, most agricultural commodity pricing in the literature [70–72] have used daily or weekly forecasts. Although using many distinct models in short-term predictions have yielded more successful results, the short-term character of these predictions appears to be a factor limiting confidence in determining long-term strategy. Recently, there has been a surge in the number of studies that take into account a wide range of variables, including product quality levels [17], past pricing [18], agricultural illnesses [19], seasonal factors [20] and climate change [21]. These studies all have the goal of making predictions based on the causal relationship between the variables. The ARIMA and machine learning-based models have been preferred, regardless of whether there is causality between the dependent and independent variables for the eleven distinct agricultural products examined in this paper. This paper has successfully forecasted commodity prices for eleven distinct agricultural items over a two-month period using three models: ARIMA, GA-LSTM, and GA-ELM. Traditional computational algorithms like ARIMA and soft computational techniques like LSTM have advantages and limitations like any other algorithm. It is known that ARIMA has the disadvantage of deviating from reality in long-term projections. During the training stages, both the ARIMA and LSTM algorithms need a significant amount of time. The ELM was developed by Huang et al. [43, 44] to reduce these drawbacks and address issues with regression, classification, clustering, feature learning, and sparse approximation. One of the most important features of ELM is that it adds a randomness feature to the ELM due to the random assignment of input weights and biases, which improves its universality and efficiency performance over other traditional models [45, 46]. A new ELM model with genetic optimization (GA) method is proposed in this paper for estimating agricultural commodities prices. The proposed methodology improves their ability to forecast agricultural commodity prices for eleven different products, including agricultural products that have recently caused food shortages and price increases in many countries around the world, including the USA and European countries. Prediction capabilities of the ELM are enhanced by automatically adjusting features such as the number of hidden nodes, layer weights, number of delays by the genetic algorithm. The GA-ELM approach has outperformed the ARIMA and GA-LSTM models, which are frequently faced in many disciplines to estimate the pricing of various products. When RMSE, MAE, and MAPE values are computed, the ELM combined with genetic algorithms appears to be quite successful in both short-term and long-term price forecasts for all eleven distinct agricultural products compared to two separate models (ARIMA and GA-LSTM). In addition, the GA-ELM, one of the approaches proposed here, seems to be quite useful because it requires less time in terms of speed factor compared to the ARIMA and GA-LSTM. It has been shown that the proposed model, unlike many studies in the literature, performs better both in closeness to reality and in determining the trends of predictions covering sixty days, making it a strong candidate for real-time agricultural commodity price forecasting applications.

5.1 Advantages of the GA-ELM model over the rival models:

- The performance of GA-ELM has a distinct advantage over other popular models, as reported in the literature [73]. It tends to generalize better, especially on complex and large data sets. In various tests, the accuracy rates of GA-ELM have been found to be statistically significantly higher than other models (*in terms of performance*).
- The GA-ELM has been successfully applied in fields such as financial forecasting [47], materials engineering [48], and industrial system [74]. This versatility demonstrates ability of the GA-ELM to adapt to various problems and its capacity to improve its performance (*in terms of application flexibility*).
- Compared to the competing models, ability of the GA-ELM to optimize the weights of the model using a genetic algorithm stands out [68]. Traditional ELM starts with random weights and optimizes these weights during the training phase. The GA-ELM, on the other hand, can find the best weight combinations more efficiently by using the genetic algorithm that increases its generalization ability (*in terms of uniqueness and validity*).
- The GA-ELM may require less training data compared to some other deep learning models [73]. This especially supports the preference of the GA-ELM in limited data situations (*in terms of usability*).
- The GA-ELM model has adaptive features managed by genetic algorithm to ensure long-term sustainability [75, 76]. In this way, it can successfully adapt to changing conditions by performing effectively according to the ELM model in terms of maintenance, updating and adaptation to data sources (Long-Term Sustainability).

In this paper, we use the RMSE measure to evaluate the robustness of the model. Agricultural commodity prices are divided into randomly selected training and validation sets in a ratio of 8:2 and various models are built and checked using different training sets. Figure 6 shows the distribution of GA-ELM, GA-LSTM and Auto-ARIMA models for eleven different agricultural commodities. Table 5 shows that the GA-ELM model is more stable compared to the GA-LSTM and Auto-ARIMA models considering the RMSE distribution. In multiple experiments conducted for the basic infrastructure of the study, the lowest RMSE and fluctuations have been observed in the GA-ELM model, which is superior in terms of accuracy and robustness. It is concluded that the predictive ability of the GA-ELM model is consistently more accurate.

In this paper, when applied to eleven different agricultural commodity prices using traditional algorithms such as ARIMA and soft calculation algorithms such as the GA-LSTM, the ARIMA stands out in some products and GA-LSTM in others, after the GA-ELM. Although both methods seem successful in short time periods, it is seen that they are below the expected success in long-term forecasts. A new model, according to the literature, is necessary to produce more accurate predictions for both the number of agricultural products and a long time period such as sixty days. In order to predict such a wide range of commodities over such a long period of time, a model called GA-ELM has been suggested in the main emphasis of this study by effectively combining the ELM and GA. When compared to the ARIMA and GA-LSTM, this model is quite successful at predicting all of the commodity values of eleven distinct agricultural products over both short and long time periods. At the same time, it has been shown that the difficulties associated with estimating agricultural commodity prices can be overcome to some extent by the ARIMA and GA-LSTM. In addition, the GA-ELM model is also very successful in terms of training time compared to the other two models. It has been demonstrated that the suggested model is in the lead both qualitatively and quantitatively, even when prices fluctuate significantly owing to global difficulties in several months of 2022. The fact that it contains daily pricing for agricultural commodity products from January 2000 to July 2022 demonstrates that this analysis is more thorough and up-to-date than earlier studies.

The proposed GA-ELM model provides higher accuracy than the ARIMA and GA-LSTM models in determining long-term price trends of agricultural products. This primarily allows policymakers to predict future price fluctuations and develop appropriate strategies. Secondly, by providing separate performance analyses for different agricultural products, the study provides policymakers with information on which products are more predictable. This information enables the development of risk management strategies. The model also predicts the future prices of agricultural products and finally, it is thought that it can contribute to managerial implications in inventory planning and supply chain. This allows policymakers to use this information effectively to maintain the balance of supply and demand. This information, which can provide guidance in dealing with price uncertainty and enable them to adapt to market conditions more effectively, can offer a significant advantage to managerial implications.

6 Conclusions and recommendation

This paper has built efficient artificial intelligence methods for effectively forecasting commodities prices considering worldwide events. The commodity prices of eleven important agricultural goods that have recently caused crises around the world have been forecasted using three independent, well-structured models. In addition to the other two proposed models, this research has created a novel model for commodity price forecasting by combining the extreme learning machine method and the genetic algorithm. The GA-ELM model has been observed to perform better at forecasting commodity prices than the GA-LSTM and ARIMA models, thus enabling successful estimation of the prices of these products, which may naturally also be strategic in the future. Despite the unexpected changes in commodity prices caused by worldwide conflicts in 2022, the GA-ELM model proposed in this study accurately predicts both the qualitative and quantitative behavior of such a large number of commodities over such a long time period. As a natural consequence of the war, it has been observed in test results and predicted results that the prices of the goods produced by the warring parties or their neighbors change the most quickly; as a result, they are among the most significant factors in shaping international relations in the short and long term. Furthermore, the richness of the number of agricultural items treated here in comparison to the literature, as well as the successful capture of commodity price trends over a reasonably long period of time, can be viewed as a major outcome of our work. The prediction that it will improve the effective management, guidance and restructuring of agricultural policies by meeting food needs that adapt to the dynamic structure of countries can be considered as an important finding of this study. However, future research should consider the impact of various objective functions in the optimization process with a genetic algorithm. This is because the objective function is the key element that drives the optimization process and therefore has a significant impact on model accuracy. It should also be noted that a fair amount of data is needed for the process if both the model parameters and the selection of input variables are estimated over different time periods. Although the results produced by all three models are very promising, it may be useful to discuss possible modifications to the models to obtain more realistic results in the future.

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Data availability The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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